

Education Disrupted

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Abstract

The computer graphics and interactive techniques industry is at the leading edge of technology and is constantly challenged by rapid advancements in technology, market shifts, and disruptive innovations. This shifting technological landscape, characterized by volatility, uncertainty, complexity, and ambiguity, at times requires educators and academic programs to adapt quickly and rethink their approaches to educational practices. This panel brings educators and industry practitioners together to identify emerging trends that disrupt education while opening new opportunities for innovation and industry collaboration. Panelists will highlight relevant pedagogical and curricular approaches and discuss how industry perspectives shape academic training to prepare learners for the evolving workforce better.

CCS Concepts

• **Applied computing** -> **Education**; • **Social and professional topics** -> **Professional topics** -> **Computing education**;

Keywords

Industry Trends, Computer Graphics Education, Pedagogical and Curricular Approaches

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1 Introduction

Rapid advancements in industry practices and disruptive technological innovations often require educators and program administrators to revisit their curriculum and modes of content delivery in academia. There is an interest in the SIGGRAPH education community to foster industry-academia partnerships to better align computer graphics education and real-world problem solving. This panel will discuss the challenges faced by educators in academic institutions to tackle the demands of the computer graphics and interactive technologies industry. Panelists will identify key disruptors and industry trends that require educators to revisit their pedagogical approaches, and share their insights on educational program models, interdisciplinary classroom learning experiences, industry perspectives on academic training, and opportunities for continued learning after graduation.

Panelists will engage in a lively conversation amongst themselves and with the audience to address a series of topics educators in art, design, and engineering are tackling to best prepare students for a long career in a rapidly changing environment. The panel structure includes an introductory round for panelists to present their perspectives, guided questions for panelists to share specific insights with examples, and time for attendees to ask questions and participate in the conversations. Panel discussions explore three major disruptive trends:

- **AI in Education** What tools and practices do students adopt in modern classrooms that force educators to change course delivery? How do we respond to using AI/GenAI tools? How are these tools viewed in the industry? How can we guide students in using these tools in a way that aligns with the industry? In what ways are these appropriate in education? What are the best sources of guidance regarding ethical and legal implications?
- **Industry Readiness & Partnership** What does the industry need, and what is academia producing? Where are the disconnects? What kind of modern skills and training lead to

long-term student career success? How do research activities and innovation fit into this workflow?

- **Learning Modalities** How does accessibility to information and student learning preferences shape the classroom experience? How does that erode the role of the academy in teaching a body of knowledge?

The panel will cover topics that reach out to a wide range of attendees in the SIGGRAPH community that include educators and students in undergraduate/graduate/K-12 education levels, and professionals seeking continued learning opportunities. The panel intends to provide attendees with resourceful insights and examples to implement in their own teaching and learning environments. The guided topics and audience interaction is expected to generate engaging conversations questioning academia's challenges and opportunities in tackling these disruptions.

2 Moderator and Panelist Bio

2.1 Moderator

Nandhini Giri is an educator, researcher, and practitioner, currently working as an assistant professor of human computer interaction and director of the experience realities lab at Purdue University. Giri studies design thinking and expertise development in computer graphics educational and interactive entertainment professional practices. Giri and her students design experiential learning systems and game-based pedagogical tools for improved student learning experiences. Her notable animated feature film credits include *Boss Baby* (2017), *Home* (2015), *Penguins of Madagascar* (2014), *Mr. Peabody and Sherman* (2014), *Rise of the Guardians* (2012), *Madagascar 3: Europe's Most Wanted* (2012).

2.2 Panelists

Ginger Alford is a Professor of Computer Science at SMU Lyle School of Engineering in Dallas, TX, where her work includes curriculum development in computer graphics, visual computing, VR/AR, and Digital Twins. Her career has balanced software engineering industry experience, intellectual property consulting and education. Committed to arts-centered engineering education at all levels, she has held Program Director roles in K-12 schools and museums overseeing development of course, curriculum, and teacher training programs, including an ongoing innovative animatronics program that embeds electronics in core middle school curriculum. She is Chair of the 2025 ACM SIGGRAPH Conference and previously chaired the SIGGRAPH education committee.

Greg Berridge has over 25 years of experience in education and the film industry, specializing in 3D content creation and virtual production. As Head of Training at Imageworks for 12 years, Greg developed the studio's internship program and academic outreach initiatives. With a Master's in Curriculum Development, the designed and led programs at VFS and remains an active educator. In 2025, they founded Adept Arts to help students transition into studio roles, providing industry-focused training in Unreal Engine and Houdini. Passionate about mentorship, Greg bridges the gap between education and professional success in film and games.

Glenn Goldman is a Professor in the Hillier College of Architecture and Design at the New Jersey Institute of Technology (NJIT) and was one of the people responsible for the creation of the School

of Art + Design, serving as its Founding Director for almost thirteen years with responsibilities for programs in digital design (including entertainment design), industrial/product design, and interior design. He is a registered architect and a licensed professional planner who has previously taught at Iowa State University and Boston Architectural Center and worked at various architecture firms. He is a former president of ACADIA (the Association for Computer-Aided Design in Architecture) and is the chair of the SIGGRAPH Education Committee.

Derek Ham is the Director of Carnegie Mellon's Entertainment Technology Center. He is an award-winning VR director, design educator, and immersive storyteller. In his research, he continues to investigate both virtual reality (VR) and augmented reality (AR) technology to find ways these tools can expand the way we work, play, and learn. His most notable projects "I Am A Man" and "Barnstormers: Determined to Win" both share a special focus on the use of virtual reality as a tool for the empathetic storytelling of African American history.

Munkhtsetseg Nandigjav Associate dean of the SCAD School of Animation and Motion, Munkhtsetseg Nandigjav has led the digital communication program as chair and associate chair and previously taught at the university for over five years. She has created designs to elevate the human experience as well as grow and nurture developing artists. Based on her background in computer graphics technology and animation, Nandigjav's diverse experience ranges from art and design to 3D production pipeline and interactive spatial experience. Nandigjav has served with ACM SIGGRAPH in a number of capacities for more than a decade, most recently chairing the SIGGRAPH 2022 conference.

Dave Shreiner has been coding and teaching computer graphics for over three decades. He's currently a Graphics Architect at Unity Technologies, and a lecturer at Sonoma State University. He's contributed to almost every aspect of computer graphics from application implementation, hardware and API design to research. He's also passionate in sharing the joy of computer graphics and interactive techniques through teaching classes (including over 40 courses at SIGGRAPH conferences), authoring books, and mentoring early career practitioners. Dave was conference chair of the 41st SIGGRAPH conference, SIGGRAPH Courses chair multiple times, and has held numerous other roles volunteering for SIGGRAPH.